Projective Geometry and Camera Calibration

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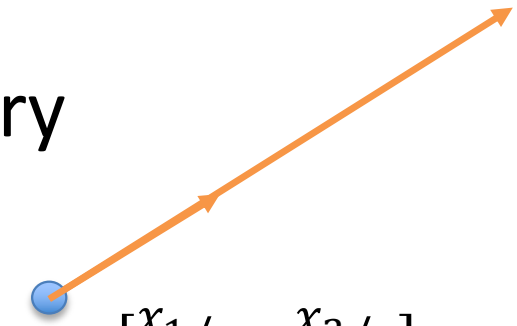
Image courtesy: https://www.etsy.com/in-en/listing/1607436850/third-eye-activation-spiritual-awakening

Review of Projective Geometry

Points and Lines in $\mathcal{P}^2$

- Points represented by: $\mathbf{x} = [x \;\; y \;\; 1]^{\mathrm{T}}$.
- Consider the line equation: $ax + by + c = 0$.
- $[a \; b \; c][x \; y \; 1]^{\mathrm{T}} = \mathbf{l}.\mathbf{x} = \mathbf{l}^{\mathrm{T}}\,\mathbf{x} = 0$, where $\mathbf{l} = [a \; b \; c]^{\mathrm{T}}$.
- Lines are represented by 3-vectors, just like points. ***Overall scale is unimportant.***
- What does $\mathbf{l}^{\mathrm{T}}\,\mathbf{x} = 0$ describe?
  - All points $\mathbf{x}$ on a fixed line $\mathbf{l}$.
  - All lines $\mathbf{l}$ passing through a fixed point $\mathbf{x}$.

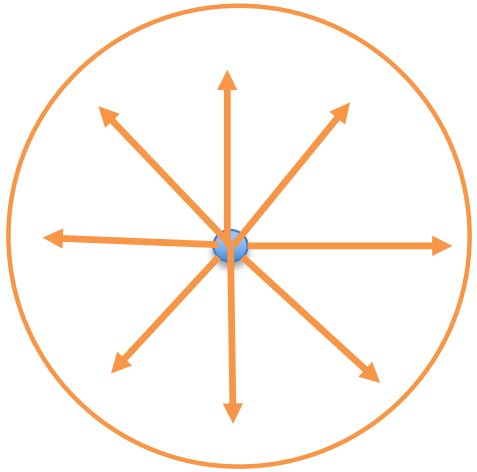
Review of Projective Geometry



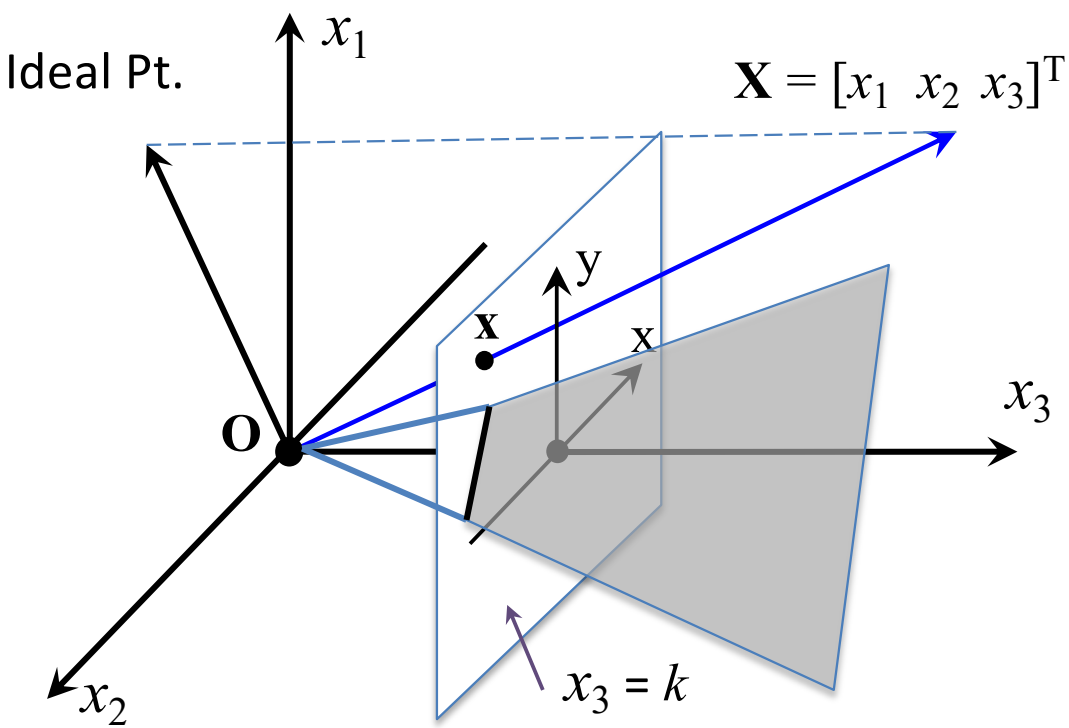
$[{}^{x_1}/_{1} \quad {}^{x_2}/_{1}]$

Points/Line at Infinity

- $\mathbf{x} = [x_1 \quad x_2 \quad x_3]^{\mathrm{T}}$ represents $[{}^{x_1}/_{x_3} \quad {}^{x_2}/_{x_3}]$
- What happens when $x_3 \rightarrow 0$?
- Becomes **point at infinity**, or **vanishing point** or **ideal point** in the direction (x_1, x_2).
- Points at infinity can be handled like any other point in projective geometry
- $[x \quad y \quad 0]^{\mathrm{T}}$ are all points at infinity on the plane.
- What do they form together?
- What is the representation of $\mathbf{l}_\infty$?
  - $\mathbf{l}_\infty = [0 \quad 0 \quad 1]^{\mathrm{T}}$



Review of Projective Geometry



Visualization of Projective Geometry of a Plane

- $\mathbf{x} = [x_1 \;\; x_2 \;\; x_3]^T$ represents rays from the origin in 3-space.
- The plane can be any cross section ⊥ to x_3.
- Ideal points are rays on the $x_3 = 0$ plane (i.e., $x_1 \; x_2$ plane).
- Lines are planes passing through the origin (and intersecting $x_3 = 0$ place).

- Line at infinity, $\mathbf{l}_\infty$, corresponds to $x_3 = 1$, x_1,x_2=0.

Review of Projective Geometry

Line Joining 2 Points

- Let p and q be points. We have: $\mathbf{l}^{\mathrm{T}}\mathbf{p} = \mathbf{l}^{\mathrm{T}}\mathbf{q} = 0$ where $\mathbf{p} = [x_1\ y_1\ 1]^T$ and $\mathbf{q} = [x_2\ y_2\ 1]^T$.

- Equation of $\mathbf{l}$: $y = y_1 + \frac{(y_2 - y_1)}{(x_2 - x_1)}(x - x_1)$

  or: $(y_2 - y_1)x - (x_2 - x_1)y + (x_2 y_1 - x_1 y_2) = 0$

  or: $\mathbf{l} = [(y_2 - y_1) \quad -(x_2 - x_1) \quad (x_2 y_1 - x_1 y_2)]^{\mathrm{T}}$

- Considering them as vectors in 3-space, we want to find a vector **l** orthogonal to both $\mathbf{p}$ and $\mathbf{q}$.

- The cross-product $\mathbf{A} \times \mathbf{B}$ is a solution. Thus, $\mathbf{l} = \mathbf{p} \times \mathbf{q}$.

$$\mathbf{p} \times \mathbf{q} = [(y_2 - y_1) \quad -(x_2 - x_1) \quad (x_2 y_1 - x_1 y_2)]^{\mathrm{T}}$$

Review of Projective Geometry

Line Joining 2 Points

- Line through (5,2) and (3,2):

$$\begin{bmatrix} i & j & k \\ 5 & 2 & 1 \\ 3 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix}$$

- $\mathbf{p} = [0 \;\; 1 \;\; -2]^{\mathrm{T}}$ i.e., $y = 2.$

- Ideal point of the line $[0 \;\; 1 \;\; -2]^{\mathrm{T}}$ is $[1 \;\; 0 \;\; 0]^{\mathrm{T}}$.
  This is same as $[0 \;\; 1 \;\; k]^{\mathrm{T}}$ for any k.

- Line joining $[3 \;\; 4 \;\; 0]^{\mathrm{T}}$ and $[2 \;\; 3 \;\; 0]^{\mathrm{T}}$ is $[0 \;\; 0 \;\; 1]^{\mathrm{T}}$ or $\mathbf{l}_{\infty}$.

Review of Projective Geometry

Point of intersection of 2 lines

- Lines $\mathbf{l}$, $\mathbf{m}$ intersect at a point $\mathbf{x}$ with $\mathbf{l}^{\mathrm{T}}\mathbf{x} = \mathbf{m}^{\mathrm{T}}\mathbf{x} = 0$.
- $\mathbf{x} = \mathbf{l} \times \mathbf{m}$.
- $\mathbf{l}$: $a_1\, x + b_1\, y + c_1 = 0$; and $\mathbf{m}$: $a_2\, x + b_2\, y + c_2 = 0$.
- $x = (b_2 c_1 - b_1 c_2)/(a_2 b_1 - a_1 b_2)$.
- $y = (a_1 c_2 - a_2 c_1)/(a_2 b_1 - a_1 b_2)$.
- $\mathbf{x} = [(b_2 c_1 - b_1 c_2)\ \ (a_1 c_2 - a_2 c_1)\ \ (a_2 b_1 - a_1 b_2)]^{\mathrm{T}} = \mathbf{l} \times \mathbf{m}$.
- Duality at work: points and lines are interchangeable.

Review of Projective Geometry

Intersection of Lines

- Intersection of x=1 and y=2:

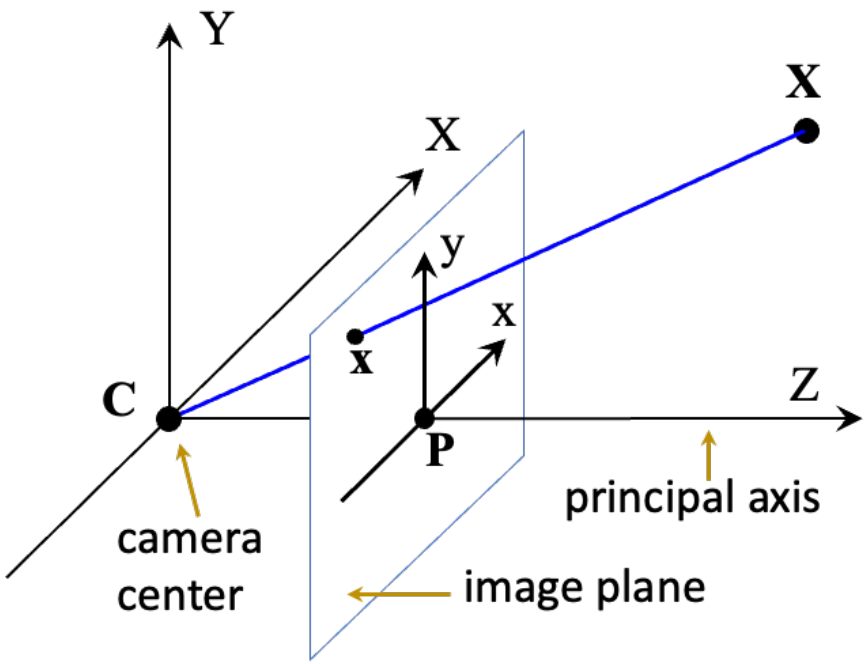
$$\begin{bmatrix} i & j & k \\ 1 & 0 & -1 \\ 0 & 1 & -2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

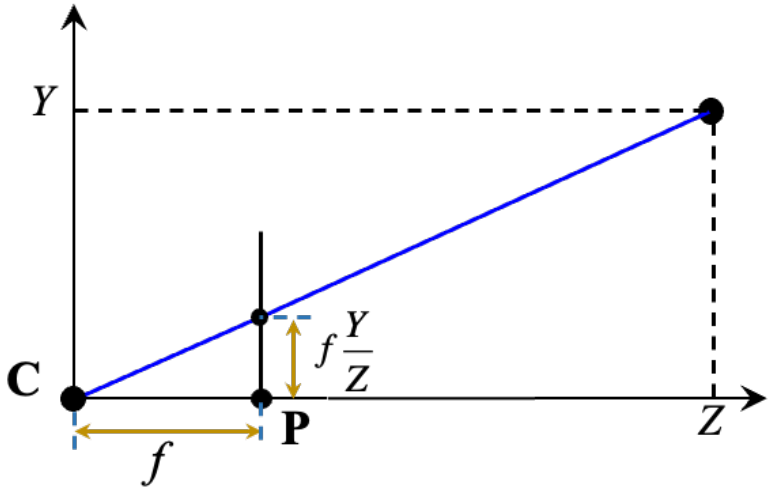
- Same as: $[1,2]$.
- Intersection of x=1 and x=2:

$$\begin{bmatrix} i & j & k \\ 1 & 0 & -1 \\ 1 & 0 & -2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

- Ideal point of the line $\mathbf{l} = [a \;\; b \;\; c]^{\mathrm{T}}$ is $[\mathrm{b} \;\; -a \;\; 0]^{\mathrm{T}}$
- This is $\mathbf{l} \times \mathbf{l}_{\infty}$, the intersection of l with line at infinity!

Recall: Geometry of Perspective Projection





$$x = f\frac{X}{Z}, \quad y = f\frac{Y}{Z}$$

Cartesian image coordinates

Recall: Mathematical Model of Perspective Projection

$$x = f\frac{X}{Z}, \quad y = f\frac{Y}{Z}$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ w \end{bmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \mathbf{PX}$$

Projection matrix (homogeneous coordinates)

A General Camera Equation

- Camera and world are related by: $$\mathbf{X_c} = \begin{bmatrix} \mathbf{R} & \mathbf{-RC} \\ \mathbf{0} & \mathbf{1} \end{bmatrix} \mathbf{X_w}$$
- 2D projection $\mathbf{x}$ of a 3D point $\mathbf{X_w}$ given by:
  - $\mathbf{x} = \mathbf{K[\ I \mid 0\ ]\ X_c} = \mathbf{K\ [\ R \mid -RC\ ]\ X_w} = \mathbf{PX_w}$
- Camera matrix $\mathbf{P} = \mathbf{[KR \mid -KRC]} = \mathbf{[M \mid p_4]}$
- Common K: $\begin{bmatrix} f & 0 & x_0 \\ 0 & f & y_0 \\ 0 & 0 & 1 \end{bmatrix}$ General K: $\begin{bmatrix} \alpha_x & s & x_0 \\ 0 & \alpha_y & y_0 \\ 0 & 0 & 1 \end{bmatrix}$

A General Camera Equation

- General projection equation in world coordinates:
  - $\mathbf{x} = \mathbf{K}\,[\mathbf{R} \mid -\mathbf{RC}]\,\mathbf{X}_w = [\mathbf{KR} \mid -\mathbf{KRC}]\,\mathbf{X}_w = [\mathbf{M} \mid \mathbf{p}_4]\,\mathbf{X}_w$
- 3x4 matrix $\mathbf{P}$ maps/projects World-C to Image-C
  - Left 3 × 3 submatrix $\mathbf{M}$ is non-singluar for finite cameras
  - Orthographic projection: left submatrix is singular
- Any 3 × 4 matrix P with non-singular left submatrix ($\mathbf{M}$) represents a camera! It can be decomposed as:
  - A non-singular upper diagonal matrix K
  - An orthonormal matrix R and a vector C with the usual meanings!!

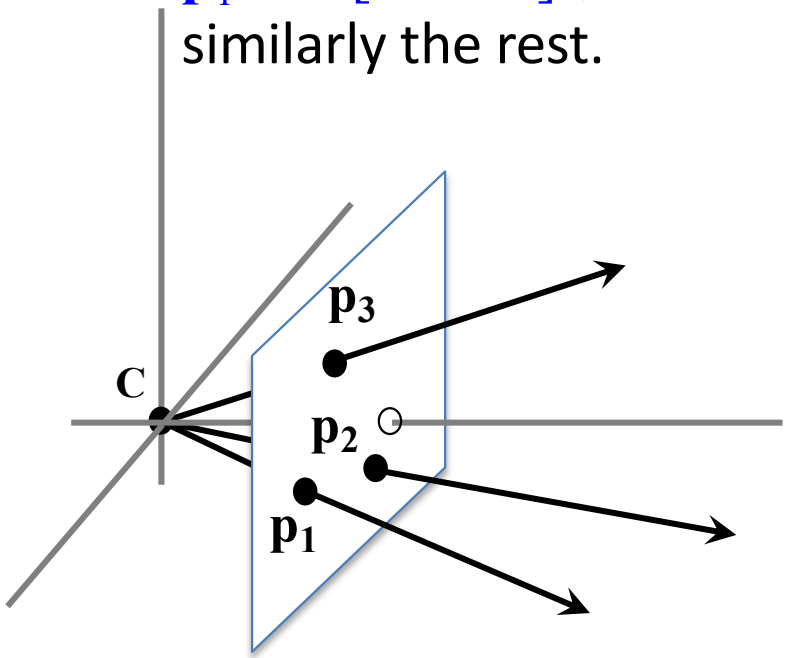
The Camera Matrix

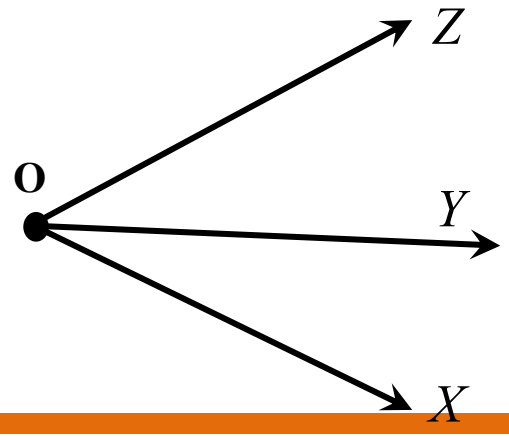
$$\mathbf{P} = \begin{bmatrix} p_{11} & \cdot\cdot & p_{14} \\ p_{21} & \cdot\cdot & p_{24} \\ p_{31} & \cdot\cdot & p_{34} \end{bmatrix} = [\, \mathbf{p}_1 \;\; \mathbf{p}_2 \;\; \mathbf{p}_3 \;\; \mathbf{p}_4 \,] = [\, \mathbf{P}^1 \;\; \mathbf{P}^2 \;\; \mathbf{P}^3 \,]^T$$

Prove the Following

$$\mathbf{P} = [\, \mathbf{p}_1 \;\; \mathbf{p}_2 \;\; \mathbf{p}_3 \;\; \mathbf{p}_4 \,] = [\, \mathbf{P}^1 \;\; \mathbf{P}^2 \;\; \mathbf{P}^3 \,]^T$$

- Show that Columns $\mathbf{p}_1$, $\mathbf{p}_2$, $\mathbf{p}_3$ are the images of vanishing points of the world X, Y and Z directions.
  - $\mathbf{p}_1 = \mathbf{P}\,[1\ 0\ 0\ 0]^T$, the image of the ideal point in X direction and similarly the rest.





Axis Points and Planes

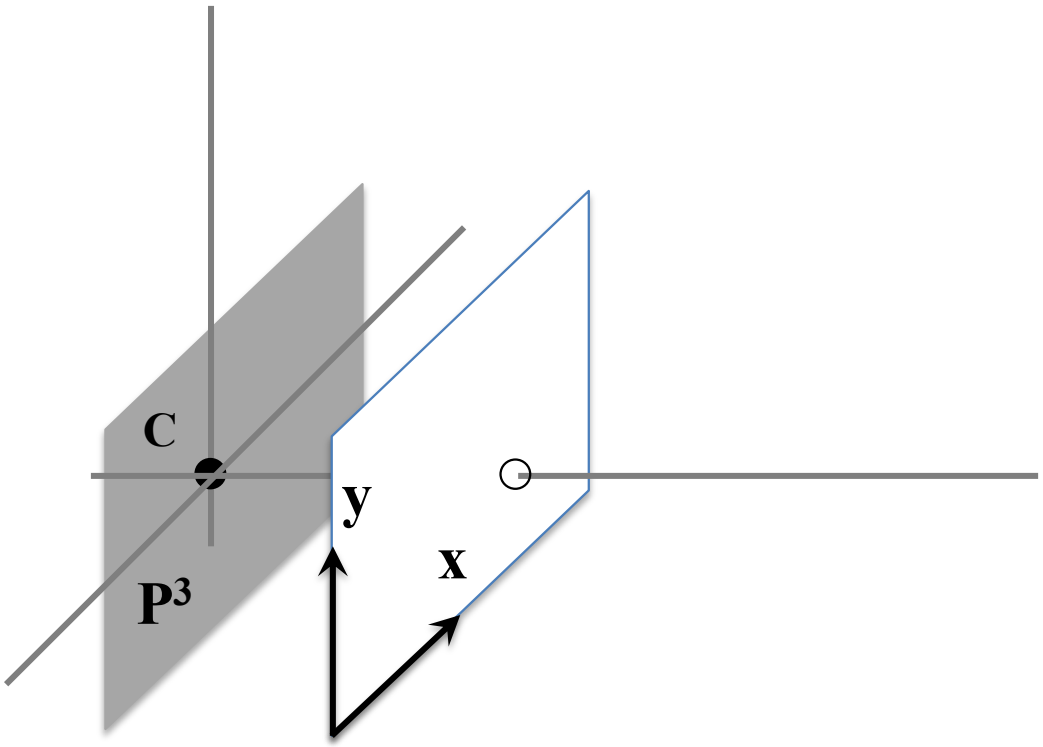
$$\mathbf{P} = [\, \mathbf{p}_1 \;\; \mathbf{p}_2 \;\; \mathbf{p}_3 \;\; \mathbf{p}_4 \,] = [\, \mathbf{P}^1 \;\; \mathbf{P}^2 \;\; \mathbf{P}^3 \,]^T$$

- 4-vector $\mathbf{C}$ with $\mathbf{PC} = \mathbf{0}$ is the camera center.
  Camera center is the only point with no projection or projects to the vector **0**, which is undefined in $\mathbf{P}^2$.

- $\mathbf{p}_4$ is the image of world origin: $\mathbf{p}_4 = \mathbf{P}\,[0\ 0\ 0\ 1]^T$.

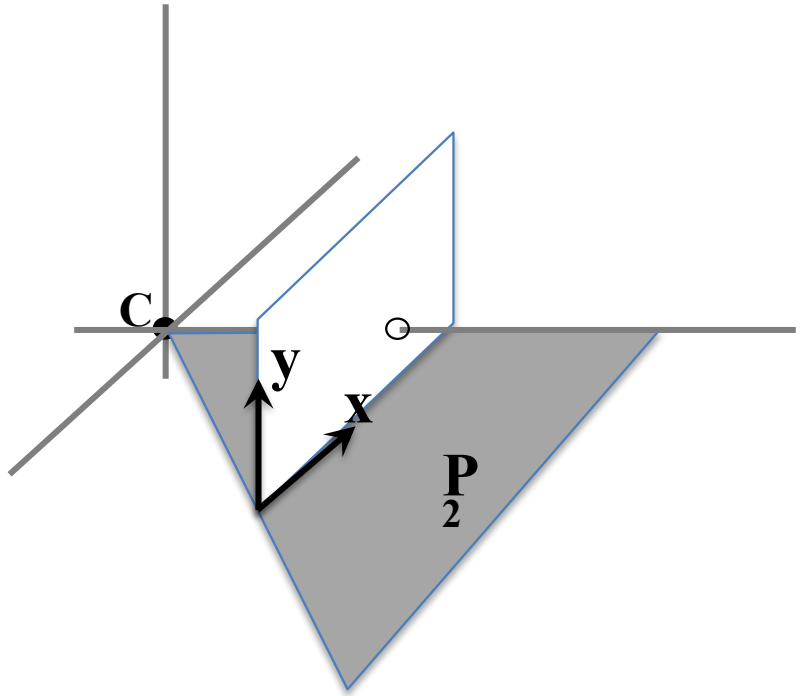
Prove the Following

- Row vector $\mathbf{P}^3$ is the principal plane.



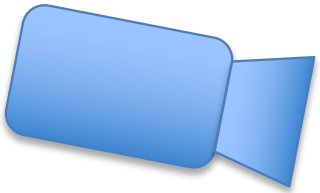
Prove the Following

- Row vectors $\mathbf{P}^1$ and $\mathbf{P}^2$ are axis planes for image Y and X axes.

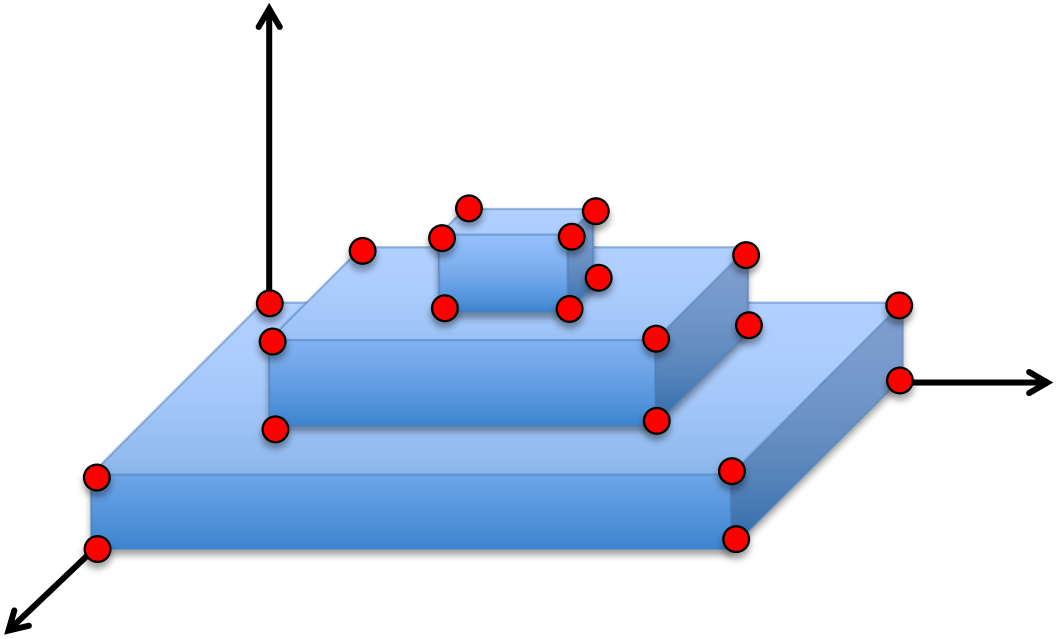


Camera Calibration

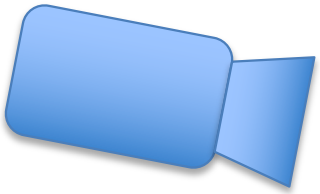
1. 3D Reference Object based calibration



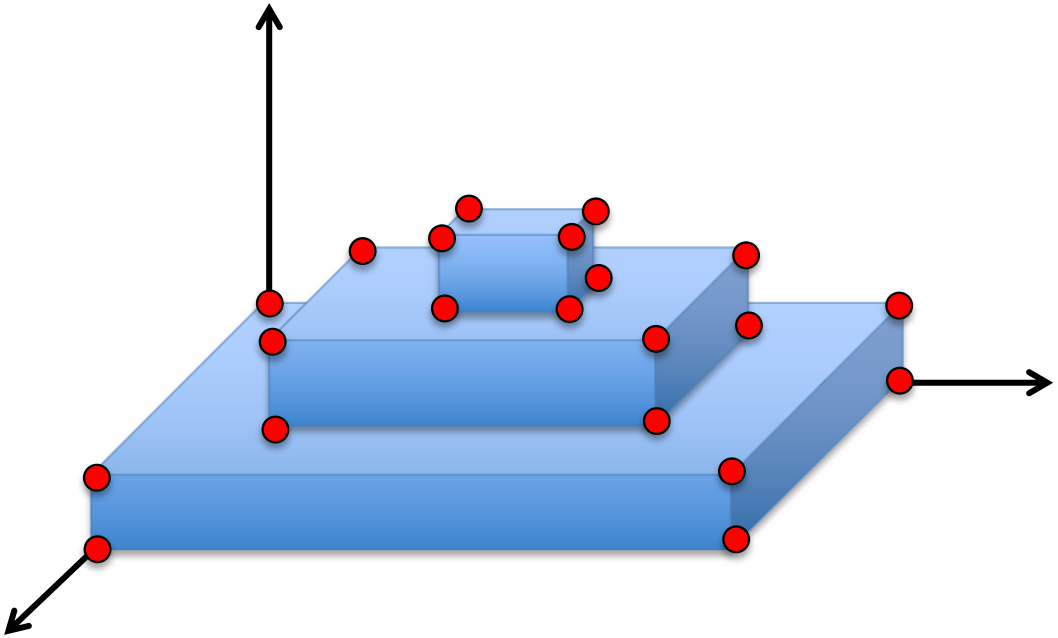
K, R, C



3D Reference Object Based Camera Calibration



K, R, C



- Given a set of world points: (X_i, Y_i, Z_i) and their corresponding image coordinates: (x_i, y_i), we can write a set of linear equations in p_{mn}, the entries of the camera matrix.

3D Reference Object Based Camera Calibration

$$\begin{bmatrix} x_i \\ y_i \\ w_i \end{bmatrix} = \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} \begin{bmatrix} X_i \\ Y_i \\ Z_i \\ 1 \end{bmatrix}$$

But we know only: $(x_i / w_i , y_i / w_i)$, Call it (u_i , v_i).

$$u_i = \frac{x_i}{w_i} = \frac{p_{11}X_i + p_{12}Y_i + p_{13}Z_i + p_{14}}{p_{31}X_i + p_{32}Y_i + p_{33}Z_i + p_{34}}$$

$$v_i = \frac{y_i}{w_i} = \frac{p_{21}X_i + p_{22}Y_i + p_{23}Z_i + p_{24}}{p_{31}X_i + p_{32}Y_i + p_{33}Z_i + p_{34}}$$

3D Reference Object Based Camera Calibration

$$\begin{bmatrix} X_i & Y_i & Z_i & 1 & 0 & 0 & 0 & 0 & -u_iX_i & -u_iY_i & -u_iZ_i & -u_i \\ 0 & 0 & 0 & 0 & X_i & Y_i & Z_i & 1 & -v_iX_i & -v_iY_i & -v_iZ_i & -v_i \end{bmatrix} \begin{bmatrix} p_{11} \\ p_{12} \\ p_{13} \\ \vdots \\ p_{34} \end{bmatrix} = [\mathbf{0}]$$

- Stack equations for all points to get $\mathbf{Gp=0}$.
- Solving this over-determined linear system of equations, we can recover the camera matrix.
- The matrix $\mathbf{P}$ can then be decomposed into the external and internal parameters: $\mathbf{K}$, $\mathbf{R}$ and $\mathbf{t}$.

Decomposing P

$$\mathbf{P} = \mathbf{K}\left[\mathbf{R} \mid \mathbf{t}\right]; \qquad \mathbf{K} = \begin{bmatrix} \alpha & \gamma & u_0 \\ 0 & \beta & v_0 \\ 0 & 0 & 1 \end{bmatrix}$$

- Let $\mathbf{P}=[\mathbf{M}\ \mathbf{p}_4]$; where $\mathbf{M}=\mathbf{KR}$, and $\mathbf{p}_4=\mathbf{Kt}$.
- $\mathbf{MM}^T = \mathbf{KRR}^T\mathbf{K}^T = \mathbf{KK}^T$. We can solve for elements of $\mathbf{K}$.
- $\mathbf{R} = \mathbf{K}^{-1}\mathbf{M}$, and $\mathbf{t} = \mathbf{K}^{-1}\mathbf{p}_4$.

Recovering K

$$\mathrm{KK^T} = \begin{bmatrix} \alpha & \gamma & u_0 \\ 0 & \beta & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \alpha & 0 & 0 \\ \gamma & \beta & 0 \\ u_0 & v_0 & 1 \end{bmatrix} = \mathrm{MM^T}$$

$$\begin{bmatrix} \alpha^2+\gamma^2+{u_0}^2 & \beta\gamma+u_0v_0 & \boxed{u_0} \\ \boxed{\beta\gamma+u_0v_0} & \boxed{\beta^2+{v_0}^2} & \boxed{v_0} \\ u_0 & v_0 & 1 \end{bmatrix} = \begin{bmatrix} mm_{11} & mm_{12} & \boxed{mm_{13}} \\ \boxed{mm_{21}} & \boxed{mm_{22}} & \boxed{mm_{23}} \\ mm_{31} & mm_{32} & mm_{33} \end{bmatrix}$$

- We get u_0 and v_0 directly.
- Compute β
- Then γ; then α.

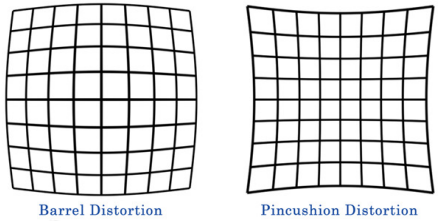
Decomposing P

- The distance metric used in the linear solution is not geometrically meaningful.
- We would like to minimize the distance between the points project by $\mathbf{P}$ and the observed points. i.e.,

$$\min_{p} \sum_{i} \|x_i - \phi(P, X_i)\|^2$$

- Can be solved by Levenberg-Marquardt (LM) algorithm.
- Use the linear solution as starting point.

Dealing with Radial Distortion



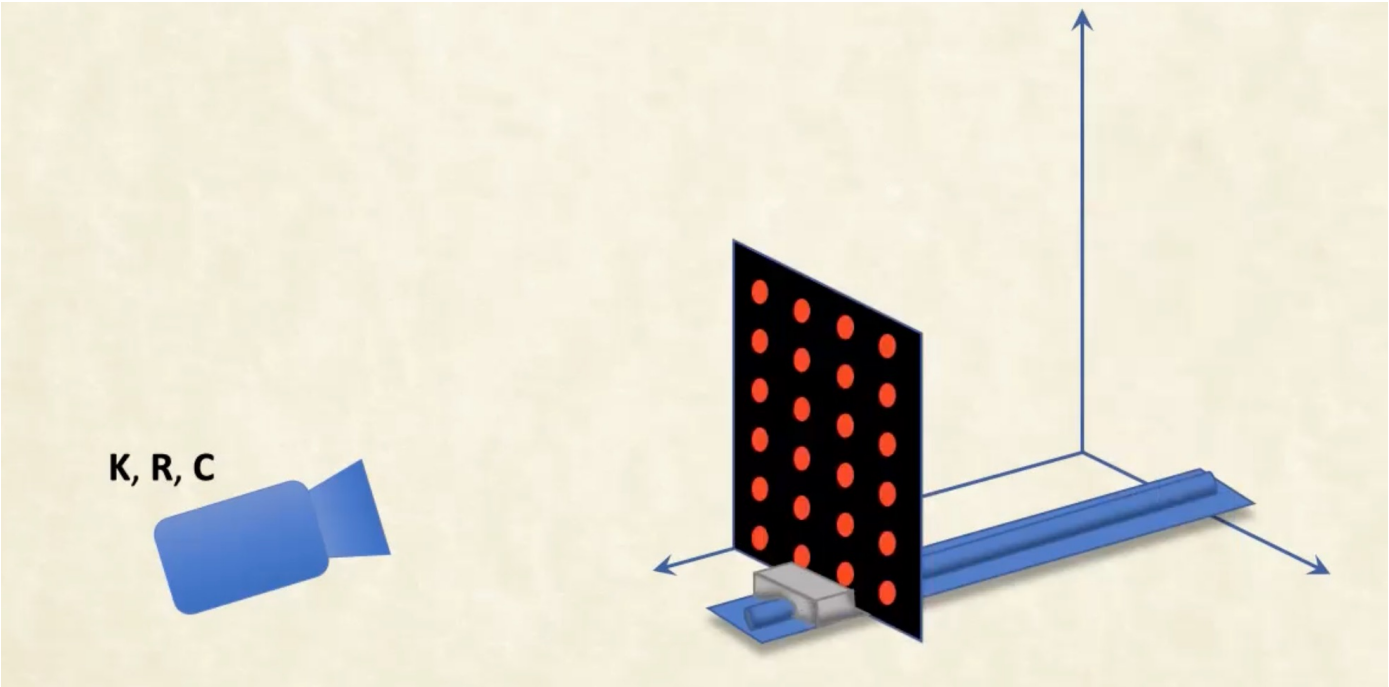
- Each pixel moves radially away from (barrel) or towards (pincushion) the image center (c).
- As a function of distance from c: ${r_c}^2 = {x_c}^2 + {y_c}^2$.
- The shift γ can be modelled as: $\gamma = 1 + k_1{r_c}^2 + k_2{r_c}^4$, where k_1 and k_2 are radial distortion parameters.
- The modified co-ordinates are:

$$\hat{x}_c = \gamma x_c$$
$$\hat{y}_c = \gamma y_c$$

- This is applied before the focal-length multiplier and center shift are applied.

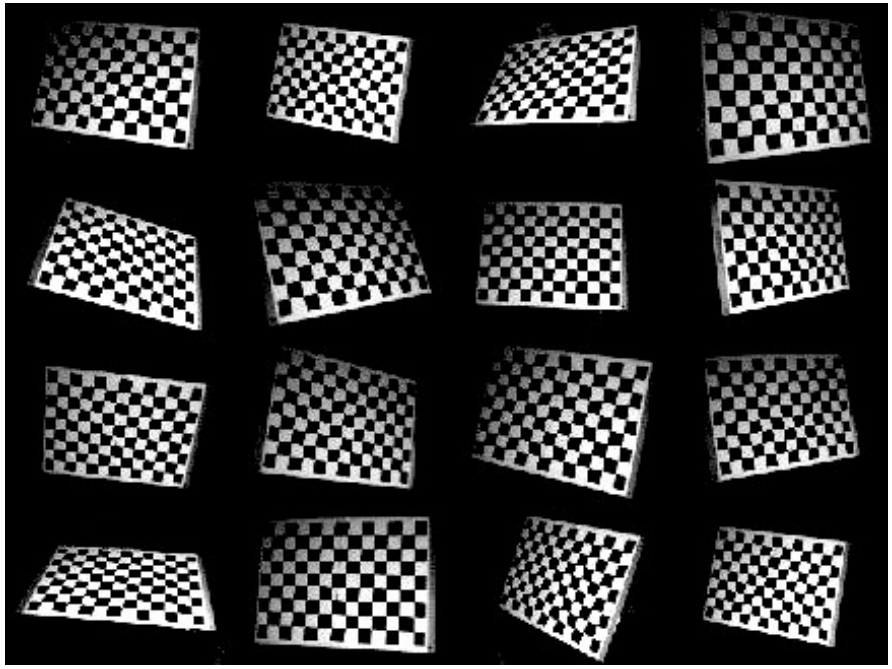
Camera Calibration

1. 3D Reference Object based calibration
2. Calibration from a precisely moving plane (R.Y. Tsai)



Camera Calibration

1. 3D Reference Object based calibration
2. Calibration from a precisely moving plane (R.Y. Tsai)
3. Calibration using a plane with unknown motion



Camera Calibration

1. 3D Reference Object based calibration
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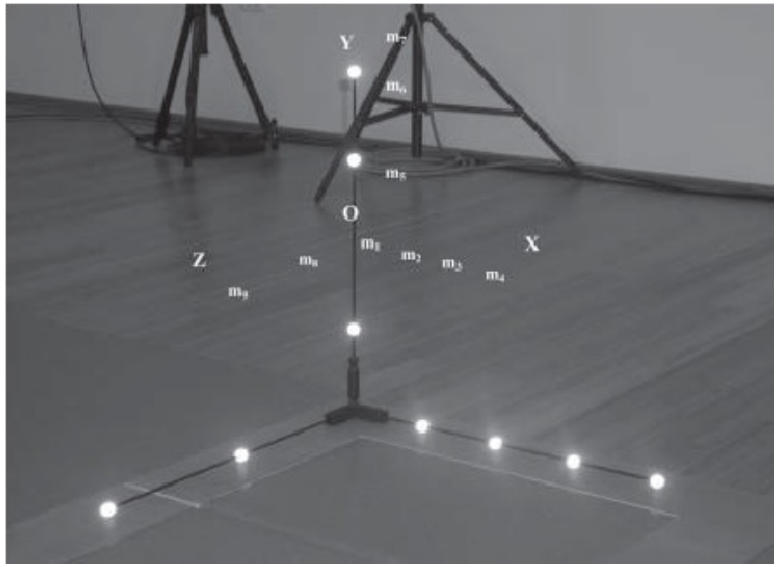
3. Calibration using a plane with unknown motion
4. Calibration from a set of collinear points that moves such that the lines passing through a fixed point



Camera Calibration

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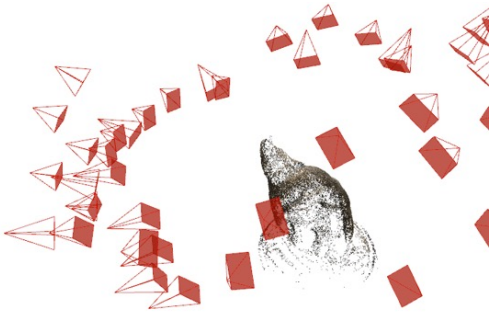
5. Self Calibration (rigid static world and point correspondences across images are available).



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- The calibration process requires fewer constraints from 1 to 5, but results become less precise.

Thank You